

## MACHINA CINEMA MODE

Machina ConsoleHub ships three operator experiences for VM access:

Mode · URL · Best for

**Machina Cinema** · `/platform/vms/:id/consolehub?mode=cinema`` · VNC, SPICE, WebRTC display ? full-screen, player-style HUD

**Machina Studio** · `/platform/vms/:id/consolehub?mode=studio`` · Serial, Shell, split panes, recovery cards, lens bar

**Mission Control wall** · `/platform/mission-control/live`` · Fleet live preview grid (max 6 concurrent thumbnails)

Public UI copy uses **Machina Cinema**, **Machina Studio**, and **Ops Shelf** ? not ?Netflix mode? or generic ?ConsoleHub tabs.?

Cinema (default display experience)

- **Hero layout:** VM canvas fills the viewport; platform sidebar and checklist are hidden.
- **Access Note pill:** Guest/NAT/SSH warnings collapse into one expandable pill (AccessNotePill).
- **Control strip:** Bottom HUD with power, scale (Fit/Fill/Native/Scroll/Stretch), screenshot, Studio toggle, Ops Shelf handle. Auto-hides after ~3.5s idle; move mouse to reveal.
- **Command palette:** ?K scoped console actions (ConsoleCommandPalette).
- **Entry points:** VM detail ?Open Cinema?, Machine Finder gallery tiles, Fleet Command Center, Spotlight, Live Preview Wall.

**Helper paths live in web/src/utils/consoleExperienceMode.ts:**

- `cinemaHubPath(vmId)` ? explicit Cinema deep link
- `studioHubPath(vmId)` ? engineer layout
- `cinemaPopoutPath(vmId)` ? centered popout window

Studio (engineer layout)

- Lens bar: Display · Serial · Shell · AI
- Full banners and recovery cards (ConsoleLoginRecoveryCard, GuestAccessBanner, ShellAccessBanner)
- Ops Shelf slide-over (refactored CommandCenterPanel) with port forwards, health, timeline
- Serial-recommended VMs auto-redirect from Cinema ? Studio

Mode persistence

**Operators who prefer Studio get their choice restored per VM:**

- Key: `localStorage['machina-console-mode:{vmId}']`
- Applied when opening `/platform/vms/:id/consolehub` **without** a mode query param
- Explicit `?mode=cinema` links (Open Cinema CTAs) always honor Cinema

Fleet surfaces

- **Gallery lens** ? Machine Finder ? Gallery ? Netflix-style rows with poster screenshots
- **Live Preview Wall** ? Mission Control launchpad ? `/platform/mission-control/live`

Graphics (VNC + SPICE)

New VMs default to graphics.type: both. Existing VMs can add/remove listeners via VM Settings ? Graphics panel or API:

- POST /api/v1/vms/{id}/graphics/add
- POST /api/v1/vms/{id}/graphics/remove

## Tests

Suite · Coverage

`web/e2e/platform-consolehub.spec.ts` · Cinema default, pill, strip idle-hide, Studio, Ops Shelf, mode restore

`web/e2e/platform-machine-finder.spec.ts` · Gallery lens ? Open Cinema

`web/e2e/platform-mission-control-live.spec.ts` · Live wall tiles

`web/src/utls/consoleExperienceMode.test.ts` · Path helpers + persistence

`web/src/utls/guestAccessHints.test.ts` · Access Note pill aggregation

## Run locally:

```
cd web && npm run build && npx playwright test e2e/platform-consolehub.spec.ts
cd web && npm test -- consoleExperienceMode
```

Enterprise console (Phase 5)

## When CONSOLEHUB\_RECORDING\_ENABLED=1 on the controller:

- Cinema shows a **Rec** badge and diagonal **Recorded · {user}** watermark
- Session history rows mark rec for recorded sessions
- **Break-glass** in Ops Shelf starts a mandatory audited + recorded session (POST .../consolehub/break-glass)
- **Spectator links**: /platform/vms/:id/consolehub?spectator={token}&session={sessionId} ? read-only view with watermark (validated via GET /api/v1/consolehub/spectator/validate)
- **Share view** (collaborative console): Cinema control strip **Share** button or Ops Shelf ? POST /api/v1/vms/{id}/consolehub/collaborate mints a time-limited read-only link for teammates
- **Session replay**: when recording is enabled, Cinema captures the VNC canvas to .webm on exit; Ops Shelf session history shows **replay** when GET /api/v1/consolehub/sessions/{id}/replay is available (CONSOLEHUB\_RECORDING\_DIR, default /var/lib/machina/console-recordings)
- **Clipboard sync**: Cinema control strip **Clipboard** button opens a panel to send laptop text into the guest via noVNC and copy guest clipboard back when the VM pushes it over VNC
- **Guest file transfer**: Ops Shelf **Send file to guest** builds an scp command (NAT or direct guest IP) after you pick a local file
- **SPICE audio**: Performance/SPICE lenses pass audio=1 to spice-html5 and enable browser autoplay; toggle in Cinema **More** menu (quality depends on guest vdaagent and spice-html5 build)
- **Multi-monitor**: when the guest framebuffer is ultra-wide (e.g. 3840×1080), Cinema infers side-by-side panels and shows **All / M1 / M2** chips in the control strip; selecting a monitor scrolls the native viewport to that region (display map shows dividers)

## RBAC from the console plan (permissions on /consolehub/plan):

Role · Power · Snapshots · Send keys

admin / operator · yes · yes · yes

viewer / readonly · no · no · no

Deferred (later enterprise)

None ? Cinema enterprise backlog is complete for v1.